Exercise 4: Find the point on the surface $xy + 2xz = 5\sqrt{5}$ that is closest to the origin.

Remark: The graph of an equation in 3 variables

$$g(x, y, z) = K = \text{const.}$$

is called a surface.

In general, it is difficult to sketch such a surface.

Solution: The distance of any point from the origin is

$$d = \sqrt{(x-0)^2 + (y-0)^2 + (z-0)^2} = \sqrt{x^2+y^2+z^2}$$

To make computations easy, we minimize d^2:

$$f(x,y,z) = d^2 = x^2 + y^2 + z^2 \qquad (1)$$

under the constraint that (x,y,z) lie on the surface:

$$g(x,y,z) = xy + 2xz = 5\sqrt{5} \qquad (2)$$

We use the method of Lagrange Multipliers:

At the point of minimum:

$$\nabla f(x,y,z) = \lambda \cdot \nabla g(x,y,z)$$

(1) (2)

$$\langle 2x, 2y, 2z \rangle = \lambda \cdot \langle y+z, x, 2x \rangle$$

We get:

$$\begin{cases} (3) \quad 2x = \lambda \cdot (y+2z) \\ 2y = \lambda \cdot x \Rightarrow y = \frac{1}{2}\lambda x \qquad (4) \\ 2z = 2\lambda x \Rightarrow z = \lambda x \qquad (5) \end{cases} \qquad (z = 2y)$$

(1) $\quad xy + 2xz = 5\sqrt{5}$ (Constraint)

Substitute (4), (5) into (3)

$$2x = \lambda\left(\frac{1}{2}\lambda x + 2\lambda x\right) = \frac{5\lambda^2 x^2}{2}$$

$$\Rightarrow \quad x = \frac{5}{4}\lambda^2 x : \quad (6)$$

We want to divide (6) by x. Note that when $x=0$, then (2) and (3) give $y = z = 0$, which by constraint (1) is not possible !!!. Therefore, $x \neq 0$, and we can divide by x in (6): $1 = \frac{5}{4}\lambda^2 \Rightarrow \lambda^2 = \frac{4}{5} \Rightarrow \lambda = \pm\frac{2}{\sqrt{5}}$

Case 1: $\boxed{\lambda = +\frac{2}{\sqrt{5}}}$ $\xRightarrow{\text{by (4)}}$ $y = \frac{1}{2}\lambda x = \frac{1}{\sqrt{5}}x$ (a)

$\xRightarrow{\text{by (5)}}$ $z = \lambda x = \frac{2}{\sqrt{5}}x = 2y$ (b)

Then the constraint (1) becomes. $[xy + 2xz = 5\sqrt{5}]$

$$x \cdot \frac{1}{\sqrt{5}}x + 2x\frac{2}{\sqrt{5}}x = 5\sqrt{5}$$

$$\frac{5}{\sqrt{5}}x^2 = 5\sqrt{5} \Rightarrow x^2 = 5 \Rightarrow \underline{\underline{x = \pm\sqrt{5}}}$$

$$\Rightarrow \underset{(a)}{y} = \frac{1}{\sqrt{5}}x = \frac{1}{\sqrt{5}}(\pm\sqrt{5}) = \pm 1$$

$$\Rightarrow z = 2y = \pm 2$$

We get 2 points: $\boxed{\begin{array}{l}(x,y,z) = (\sqrt{5}, 1, 2) \\ (x,y,z) = (-\sqrt{5}, -1, -2)\end{array}}$

<u>Case 2:</u> $\boxed{\lambda = -\frac{2}{\sqrt{5}}}$ $\overset{(4)}{\Longrightarrow}$ $y = \frac{1}{2}\lambda x = -\frac{1}{\sqrt{5}}x$

$$z = 2y = -\frac{2}{\sqrt{5}}x$$

Plug into the Constraint (1):

$$x\cdot\left(-\frac{1}{\sqrt{5}}x\right) + 2x\left(-\frac{2}{\sqrt{5}}x\right) = 5\sqrt{5}$$

$$5x^2 = -25 \quad \times \text{ impossible}$$

So there are <u>no</u> x, y, z for $\lambda = -\frac{1}{\sqrt{5}}x$.

Plug the two points $(\sqrt{5},1,2)$ and $(-\sqrt{5},-1,-2)$ into $f(x,y,z) = z^2$

$$f(\sqrt{5},1,2) = \sqrt{5}^2 + 1^2 + 2^2 = 10$$

$$f(-\sqrt{5},-1,-2) = 10$$

Because there must be a smallest d^2,

So $f(\sqrt{5},1,2) = f(-\sqrt{5},-1,-2) = (\pm\sqrt{5})^2 + (\pm 1)^2 + (\pm 1)^2$

$$= \underline{\underline{10}}$$

<u>Answer:</u> The minimum distance of $\sqrt{10}$ is occurs at the two points $(-5,-1,-2)$ and $(5,1,2)$ on the surface

$$\underline{\underline{xy + 2xz = 5\sqrt{5}}}$$

Remark: In the method of Lagrange multipliers, one can have several constraints. For example, to maximize/minimize a function $f(x, y, z)$ under the two constraints

$$g(x, y, z) = k \qquad (1) \qquad \leftarrow \text{ there are}$$

and

$$h(x, y, z) = c \qquad (2) \qquad \leftarrow \text{ two constraints.}$$

One has to solve

$$\nabla f(x, y, z) \;=\; \lambda \nabla g(x, y, z) \;+ \mu \nabla h(x, y, z)$$

Under constraints (1) + (2).

Example 10: Find the maximum value of the function

$$f(x,y,z) = x + 2y + 3z$$

on the curve of intersection of the

plane $\quad x - y + z = 1$ ← plane with normal vector $\vec{n} = \langle 1, -1, 1 \rangle$

and the cylinder $\quad x^2 + y^2 = 1,$

(in 3 dimensions, this is a cylinder)

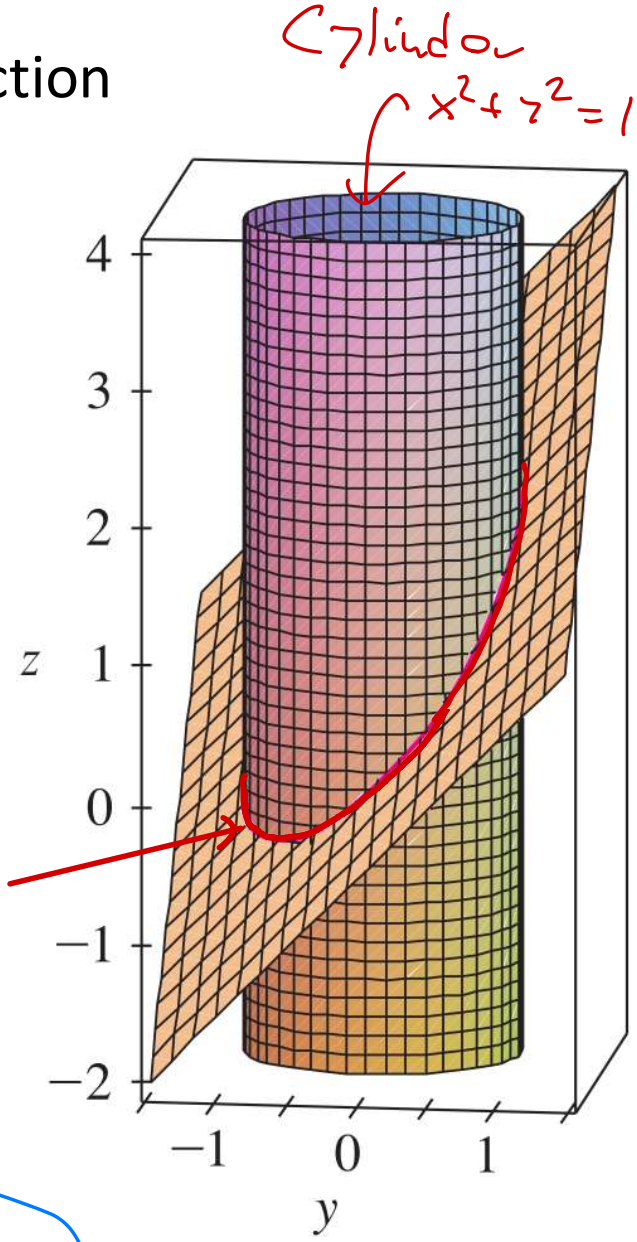
Solution: We maximize the function

$$f(x,y,z) = x + 2y + 3z$$

subject to the constraints

(1) $\quad g(x,y,z) = x - y + z = 1$ and

(2) $\quad h(x,y,z) = x^2 + y^2 = 1$



1. By the method of Lagrange multipliers, we must solve

$$\nabla f(x,y,z) = \lambda \nabla g(x,y,z) + \mu \nabla h(x,y,z)$$

$$\langle 1, 2, 3 \rangle = \lambda \cdot \langle 1, -1, 1 \rangle + \mu \langle 2x, 2y, 0 \rangle$$

$$\begin{cases} (a) & 1 = \lambda + 2\mu x \\ (b) & 2 = -\lambda + 2\mu y \\ & 3 = \lambda \end{cases} \Rightarrow \boxed{\lambda = 3}$$

$$\left.\begin{array}{ll} (1) & x - y + z = 1 \\ (2) & x^2 + y^2 = 1 \end{array}\right\} \text{Constraints}$$

Substitute $\lambda = 3$ into (a) + (b):

$$1 = 3 + 2\mu x \Rightarrow 2\mu x = -2 \Rightarrow \mu x = -1 \quad (c)$$

$$2 = -3 + 2\mu y \Rightarrow 2\mu y = 5 \Rightarrow \mu y = \frac{5}{2} \quad (d)$$

we want to divide by μ.

if $\mu = 0 \underset{(a)}{\Rightarrow} \lambda = 1$ $\times$ impossible because $\lambda = 3$.

So in (c), (d) we can divide by μ:

$$x = -\frac{1}{\mu} \qquad y = \frac{5}{2} \cdot \frac{1}{\mu} \qquad (e)$$

Substitute into constraint (2):

$$\left(-\frac{1}{\mu}\right)^2 + \left(\frac{5}{2} \cdot \frac{1}{\mu}\right)^2 = 1$$

$$\frac{29}{4} = \mu^2 \Rightarrow \boxed{\mu = \pm \frac{\sqrt{29}}{2}}$$

By (e):

$\left(\mu = \frac{+\sqrt{29}}{2}\right)$ $\quad x = \underline{\underline{\frac{-2}{\sqrt{29}}}}$ and $y = \frac{5}{2} \cdot \frac{2}{\sqrt{29}} = \underline{\underline{\frac{5}{\sqrt{29}}}}$

To find z, substitute into the (so far unused) constraint C1:

$$x - y + z = 1 \Rightarrow z = 1 - x + y$$

$$\Rightarrow z = 1 - \left(-\frac{2}{\sqrt{29}}\right) + \frac{5}{\sqrt{29}} = \underline{\underline{1 + \frac{7}{\sqrt{29}}}}$$

$\left(\mu = -\frac{\sqrt{29}}{2}\right)$ $\quad x = -\frac{1}{\mu} = \underline{\underline{\frac{2}{\sqrt{29}}}}$, $\quad y = \frac{5}{2} \cdot \frac{1}{\mu} = \frac{5}{2} \cdot \left(-\frac{2}{\sqrt{29}}\right) = \underline{\underline{-\frac{5}{\sqrt{29}}}}$

$$\Rightarrow z = 1 - x + y = 1 - \frac{2}{\sqrt{29}} + \left(-\frac{5}{27}\right) = \underline{\underline{1 - \frac{7}{\sqrt{29}}}}$$

We have found two points:

$$\left(-\frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}}, 1+\frac{7}{\sqrt{29}}\right) \text{ and } \left(\frac{2}{\sqrt{29}}, \frac{-5}{29}, 1-\frac{7}{\sqrt{29}}\right)$$

We substitute thes into $f(x,y,z) = x+2y+3z$:

$$f\left(-\frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}}, 1+\frac{7}{\sqrt{29}}\right) = -\frac{2}{\sqrt{29}} + 2\cdot\frac{5}{\sqrt{29}} + 3 + \frac{21}{\sqrt{29}} = 3 + \frac{29}{\sqrt{29}}$$

$$= \underline{\underline{3+\sqrt{29}}}$$

$$f\left(\frac{2}{\sqrt{29}}, -\frac{5}{\sqrt{29}}, 1-\frac{7}{\sqrt{29}}\right) = \frac{2}{\sqrt{29}} + 2\left(-\frac{5}{\sqrt{29}}\right) + 3 - \frac{21}{\sqrt{29}} = 3 - \frac{29}{\sqrt{29}}$$

max value

$$= \underline{\underline{3-\sqrt{29}}}$$

min value

Exercise 5: Find the maximum and minimum values of the function

$$f(x,y,z,t) = x + y + z + t \qquad (1)$$

on the hypersphere $$x^2 + y^2 + z^2 + t^2 = 1 \qquad (2)$$

Solution: By the method of Lagrange multipliers,

Solve $\nabla f(x,y,z,t) = \lambda \cdot \nabla g(x,y,z,t)$

under constraint (2)

$$\langle 1, 1, 1, 1 \rangle = \lambda \cdot \langle 2x, 2y, 2z, 2t \rangle$$

We must solve

$$\begin{cases} (a) & 1 = 2\lambda x & \Rightarrow x = \frac{1}{2\lambda} \\ (b) & 1 = 2\lambda y & \Rightarrow y = \frac{1}{2\lambda} \\ (c) & 1 = 2\lambda z & \Rightarrow z = \frac{1}{2\lambda} \\ (d) & 1 = 2\lambda t & \Rightarrow t = \frac{1}{2\lambda} \end{cases}$$

($\lambda \neq 0$. Because when $\lambda = 0 \Rightarrow x = y = z = t = 0$ and then constraint (2) gives $0 = 1$ ✗)

Constraint $\rightarrow$ $x^2 + y^2 + z^2 + t^2 = 1$

Substitute (a) (b) (c) (d) into the constraint:

$$\left(\frac{1}{2\lambda}\right)^2 + \left(\frac{1}{2\lambda}\right)^2 + \left(\frac{1}{2\lambda}\right)^2 + \left(\frac{1}{2\lambda}\right)^2 = 1$$

$$\frac{4}{4\lambda^2} = 1 \Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$$

use (a) – (d):

$\lambda = 1$: $x = y = z = t = \frac{1}{2}$, $f\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \underline{\underline{2}}$

$\lambda = -1$: $x = y = z = t = -\frac{1}{2}$, $f\left(-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right) = -\frac{1}{2} - \frac{1}{2} - \frac{1}{2} - \frac{1}{2} = \underline{\underline{-2}}$

Max value

Min value.

Assignment 2: (Due Wednesday, 16. July)

Use Lagrange multipliers to find the maximum and minimum values of the function subject to the given constraint(s).

1. $f(x,y) = x^2 y$, constraint: $g(x,y) = x^2 + 2y^2 = 6$

   (You should get 6 points (x,y) !)

2. $f(x,y,z) = x^2 y^2 z^2$, constraint: $x^2 + y^2 + z^2 = 1$

3. $f(x,y,z) = 3x - y - 3z$

   2 constraints: $x + y - z = 0$ and $x^2 + 2z^2 = 1$

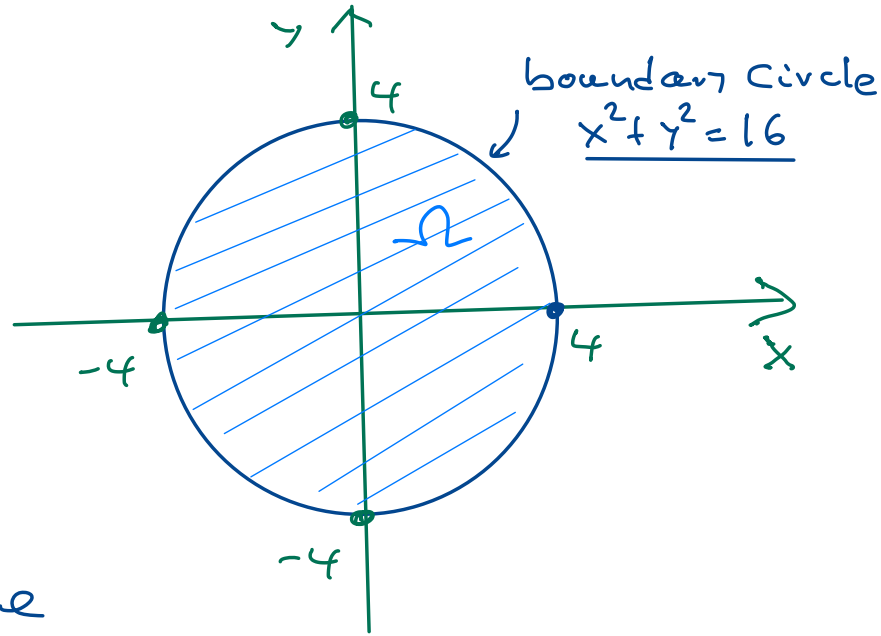
There will be a quiz on Wednesday, July 16.

Topic: Local max/min of functions of 2 variables.

Exercise 6: Find the maximum and minimum values of the function

(1) $f(x,y) = 2x^2 + 3y^2 - 4x - 5$ on the disk $x^2 + y^2 \le 16.$

(use Lagrange multipliers for the boundary)



Solution:

Note: This disk is a closed and bounded set, and f is continuous.

$\Rightarrow$ By the Extreme Value Theorem, f has a max/min on the disk

We proceed as follows:

(1) Find the critical points of $f(x,y)$ inside the disk.

(2) Find the points on the boundary circle where $f(x,y)$ is largest / smallest

③ Compare the values $f(x,y)$ from the points obtained in ① and ②.

Step ①: Find the critical points:

$$f(x,y) = 2x^2 + 3y^2 - 4x - 5 \quad (1)$$

$$f_x(x,y) = 4x - 4 \qquad f_y(x,y) = 6y$$

$f_x(x,y) = 0$ when $4x - 4 = 0 \Rightarrow 4x = 4 \Rightarrow x = 1$

$f_y(x,y) = 0$ when $6y = 0 \Rightarrow y = 0$

One critical point: $(1,0)$

Then $f(1,0) \underset{(1)}{=} 2 + 0 - 4 - 5 = -7$ (i)

Step ②: Find max/min of f on the boundary circle.

$$\begin{cases} \text{Maximize/Minimize} & f(x,y) = 2x^2 + 3y^2 - 4x - 5 \quad (1) \\ \text{Constraint:} & g(x,y) = x^2 + y^2 = 16 \quad (2) \end{cases}$$

Use Lagrange Multipliers:

$$\nabla f(x,y) = \lambda \nabla g(x,y)$$

$$\langle 4x-4, 6y \rangle = \lambda \langle 2x, 2y \rangle$$

$$\begin{cases} (a) & 4x-4 = 2\lambda x \\ (b) & 6y = 2\lambda y \\ (c) & x^2 + y^2 = 16 \end{cases}$$

By (b), $2\lambda y - 6y = 0 \Rightarrow 2y(\lambda - 3) = 0$

$y = 0$ or $\lambda = 3$

Case 1: $y = 0$. (Then λ is arbitrary)

By (c) $x^2 = 16 - y^2 = 16 \Rightarrow x = \pm 4$

We have obtained 2 points: $(4, 0)$ and $(-4, 0)$

$x \quad y \qquad x \quad y$

By (1), $f(4, 0) = 2 \cdot 4^2 + 3 \cdot 0 - 4 \cdot 4 - 5 = 11$ – (ii)

$f(-4, 0) = 2 \cdot (-4)^2 + 3 \cdot 0 - 4 \cdot (-4) - 5 = 43$ – (iii)

<u>Case 2:</u> <u>$\lambda = 3$.</u> Then (a) becomes

$$4x - 4 = 2 \cdot 3 \cdot x \Rightarrow 2x = -4 \Rightarrow \underline{\underline{x = -2}}$$

and by (c):

$$y^2 = 16 - x^2 = 16 - (-2)^2 = 12 \Rightarrow y = \pm\sqrt{12} = \underline{\underline{\pm 2\sqrt{3}}}$$

We have obtained another 2 points: $(-2, \pm 2\sqrt{3})$

By (1), $f(-2, \pm\sqrt{3}) \underset{(1)}{=} 2 \cdot (-2)^2 + 3 \cdot 12 - 4(-2) - 5 = 47$

(iv)

Step ③: Compare the values of f in (i)–(iv). We see:

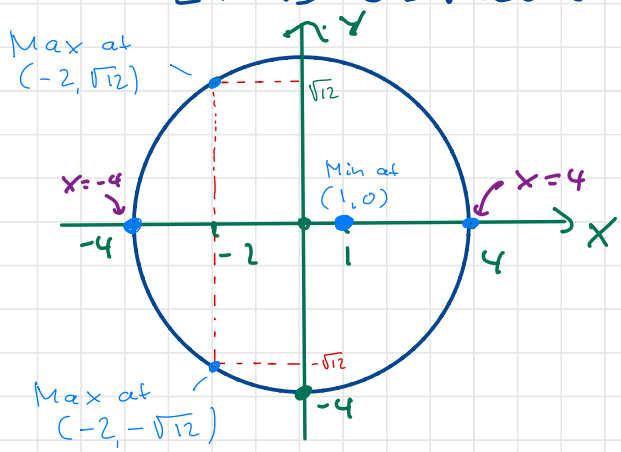
↳ $-7, 43, 11, 47$

The <u>max.</u> value of f is $\underline{\underline{47}}$.

It is obtained on the <u>boundary circle</u> at $(x,y) = \underline{\underline{(-2, 2\sqrt{3})}}$ and $(x,y) = \underline{\underline{(-2, -2\sqrt{3})}}$

The <u>min.</u> value of f is $\underline{\underline{-7}}$.

It is obtained inside the disk at $(x,y) = \underline{\underline{(1,0)}}$.



Remark: There is another method for step (2):

$$f(x,y) = 2x^2 + 3y^2 - 4y - 5 \qquad (1)$$

$$x^2 + y^2 = 16 \qquad (2)$$

By (2),

$$y^2 = 16 - x^2. \qquad (3)$$

Substitute into (1)

$$h(x) = f(x, 16 - x^2) = 2x^2 + 3[16 - x^2] - 4x - 5$$

(4) $\boxed{h(x) = -x^2 - 4x + 43}$ $\boxed{-4 \le x \le 4}$

on the boundary disk

(a) Find the critical points of h

$h'(x) = -2x - 4$ $\quad h'(x) = 0$ when $-2x - 4 = 0$

$2x = -4 \Rightarrow \underline{\underline{x = -2}}$

When $x=-2 \Rightarrow y^2 = 16 - x^2 = 16 - 2^2 = 12$

$\Rightarrow y = \pm\sqrt{12} = \pm 2\sqrt{3}$

$h(-2) = f(-2, \pm 2\sqrt{3}) \underset{(4)}{=} 47$ ← this is (iv)

(5) We must also check at the endpoints $x=-4,\ x=4$

When $x = \pm 4 \Rightarrow y^2 = 16 - x^2 = 16 - (\pm 4)^2 = 0$

$\Rightarrow y = 0$

this is (ii)

$h(4) = f(4, 0) \underset{(4)}{=} -16 - 16 + 43 = 11$

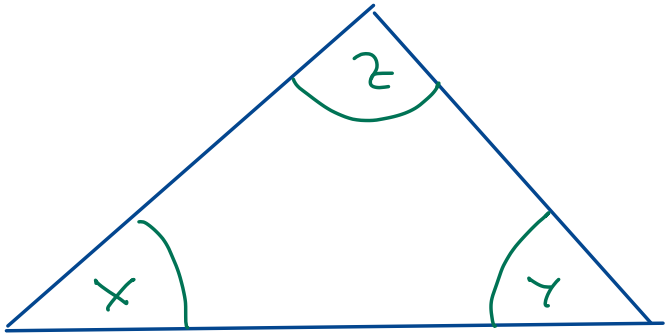
this is (iii)

$h(-4) = f(-4, 0) \underset{(4)}{=} -16 + 16 + 43 = 43$

Now compare (i)–(iv) again

Exercise 7: Let x, y, z denote the angles of an arbitrary triangle.

Find the angles that maximize the product: $\sin x \sin y \sin z$



Maximize $f(x,y,z) = \sin x \cdot \sin y \cdot \sin z$

Constraint: $g(x,y,z) = x + y + z = 180^\circ = \pi$

(also: $0 < x, y, z < \pi$)

Use Lagrange Multipliers:

$$\nabla f(x,y,z) = \lambda \nabla g(x,y,z)$$

$$\langle \cos x \cdot \sin y \cdot \sin z, \sin x \cdot \cos y \cdot \sin z, \sin x \cdot \sin y \cdot \cos z \rangle$$

$$= \lambda \cdot \langle 1, 1, 1 \rangle$$

We must solve

$$\begin{cases} \cos x \cdot \sin y \cdot \sin z = \lambda & (a) \\ \sin x \cdot \cos y \cdot \sin z = \lambda & (b) \\ \sin x \cdot \sin y \cdot \cos z = \lambda & (c) \\ x + y + z = \pi & \end{cases}$$

$(a) = (b)$: $\cos x \cdot \sin y \cdot \sin z = \sin x \cdot \cos y \cdot \sin z$

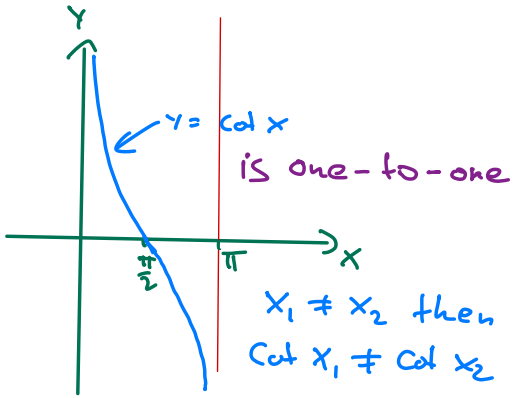
[$\sin x, \sin y, \sin z \neq 0$ because $x, y, z \neq 0, \neq \pi$]

We can divide by $\sin z \neq 0$:

$$\cos x \cdot \sin y = \sin x \cdot \cos y$$

$$\frac{\cos x}{\sin x} = \frac{\cos y}{\sin y}$$

$$\cot x = \cot y$$



Because $0 < x, y < \pi$ and $\cot x$ is one-to-one on $(0, \pi) \Rightarrow \boxed{x = y}$

$(S) = (C)$ $\quad \cancel{\sin x} \cos y \sin z = \cancel{\sin x} \sin y \cos z$

$$\frac{\cos y}{\sin y} = \frac{\cos z}{\sin z}$$

$$\cot y = \cot z \implies \boxed{y = z}$$

We have found: $x = y = z$

Because $x + y + z = \pi \implies \underline{\underline{x = y = z = \frac{\pi}{3}}}$ $(= 60^\circ)$

Assignment 2: (Due Wednesday, 16. July)

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   (You should get 6 points (x,y) !)

2. $f(x,y,z) = x^2 y^2 z^2$, constraint: $x^2 + y^2 + z^2 = 1$

3. $f(x,y,z) = 3x - y - 3z$

   2 constraints: $x + y - z = 0$ and $x^2 + 2z^2 = 1$

There will be a quiz on Wednesday, July 16.

Topic: Local max/min of functions of 2 variables.